

Recession & Bear Market Indicator V3

A Rules-Based, Persistence-Filtered Composite Signal
for Systematic Equity Underweighting

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Abstract. We develop a monthly binary indicator designed to identify, in real or near-real time, regimes of seriously negative economic and market conditions that warrant a full underweight of equities in a multi-asset portfolio. The indicator combines four independently gated branches — macro deterioration, credit market stress, rate/valuation bear, and trend-plus-macro alignment — operating under a strict AND logic within each branch and OR logic across branches. A four-state persistence filter prevents whipsaw. Backtested over 660 monthly observations from January 1970 to December 2024, the indicator achieves an 80.4% NBER recession hit rate, 9.3% false positive rate, 70.1% equity precision, and 3.3 signal reversals per decade, with the S&P 500 returning +9.5% p.a. annualised during inactive periods versus +4.0% during active periods.

Table of Contents

1. Motivation and Investment Objective.....	3
2. Design Philosophy and Architecture Overview.....	4
3. Data Sources and Construction.....	5
4. Score Computation.....	6
4.1 Market sub-score (M1–M4).....	6
4.2 Macro sub-score (A1–A7).....	7
4.3 Composite score.....	7
5. Entry Branches — Decision Logic.....	8
5.1 Branch A: macro deterioration.....	8
5.2 Branch B: credit/market stress.....	9
5.3 Branch C: rate/valuation bear.....	9
5.4 Branch D: trend + full macro alignment.....	10
5.5 Branch interaction and OR combination.....	10
6. Persistence Filter — Four-State Machine.....	11
7. Exit Conditions.....	12
7.1 Standard exit (3-month confirmation).....	12
7.2 Accelerated exit (immediate override).....	12
7.3 Accepted residual lags.....	13
8. Early Warning Layer.....	13
9. Backtest Results and Performance Attribution.....	14
10. Redundancy Analysis.....	16
11. Robustness Assessment.....	17
11.1 Block bootstrap.....	17
11.2 Leave-one-recession-out.....	18
11.3 Threshold sensitivity.....	18
12. Limitations and Known Weaknesses.....	19
13. Implementation Guidance.....	20
Appendix A Data Provenance Manifest.....	21
Appendix B Complete Threshold Table.....	21

1. Motivation and Investment Objective

Multi-asset portfolio managers face a recurring challenge: identifying, with sufficient lead time and without excessive false positives, those regimes in which a full underweight of equities is warranted. The challenge has two symmetric costs. A false negative — remaining at benchmark equity weight during a serious bear market or recession — directly transmits significant drawdown into the portfolio. A false positive — implementing an underweight during a mere correction — forgoes the subsequent recovery and reduces long-run compounding. The institutional investor with liability-matching constraints or explicit drawdown budgets cannot tolerate either failure mode in excess.

Existing approaches divide broadly into two families. Pure macro indicators — the NBER recession flag, the Conference Board LEI, the Sahm Rule — offer economically grounded signals but are systematically late (the NBER dates recessions with a 12–18 month lag) and ignore structural bear markets that occur without formal recession (2002 technology bear, 2022 inflation-driven repricing). Pure market indicators — moving average crossovers, drawdown triggers, VIX spikes — are fast but noisy, generating multiple false positives per decade during normal corrections from which the market recovers quickly.

This paper documents a third approach: a composite indicator that requires simultaneous confirmation from both market stress conditions and macro deterioration signals. The core architectural constraint — *no branch fires on market stress alone, and no branch fires on macro weakness alone* — defines it as a strict subset of seriously negative economic and market conditions, rather than a union of either alone. The four-state persistence filter further ensures the signal can only change state when evidence has accumulated over multiple months, preventing single-observation reversals.

Investment objective. When the indicator is active (signal = 1):

- Implement a full underweight of equities in the strategic asset allocation.
- Maintain the underweight until all exit conditions are simultaneously satisfied.
- During the pre-signal early warning phase (composite ≥ 3 , rising), move to neutral.
- When the signal deactivates, return to neutral first before moving to overweight.

2. Design Philosophy and Architecture Overview

The design is governed by five principles derived from a forensic examination of every false positive and false negative in the backtest.

Principle 1: AND logic within branches, OR logic across branches. Each entry branch requires simultaneous satisfaction of at least two conditions from different domains — a market stress condition and a macro confirmation condition. This joint requirement eliminates the 1998 LTCM crisis (credit stress but ISM=54), the 2011 European sovereign crisis (HY widening but ISM=55), and the 2019 Q4 correction (HY elevated but ISM=53) — all episodes where market fear was high but the underlying economy was not seriously deteriorating. The four branches are then OR-combined to capture different regime archetypes.

Principle 2: Strict subsetting. The indicator is designed to be a strict subset of seriously negative conditions, not a superset. It deliberately accepts false negatives (missing parts of short recessions) to avoid false positives. The 1987 Black Monday episode — the only major bear market with no macro deterioration whatsoever — is excluded from the target universe because no leading indicator can identify it without generating intolerable noise across every subsequent correction.

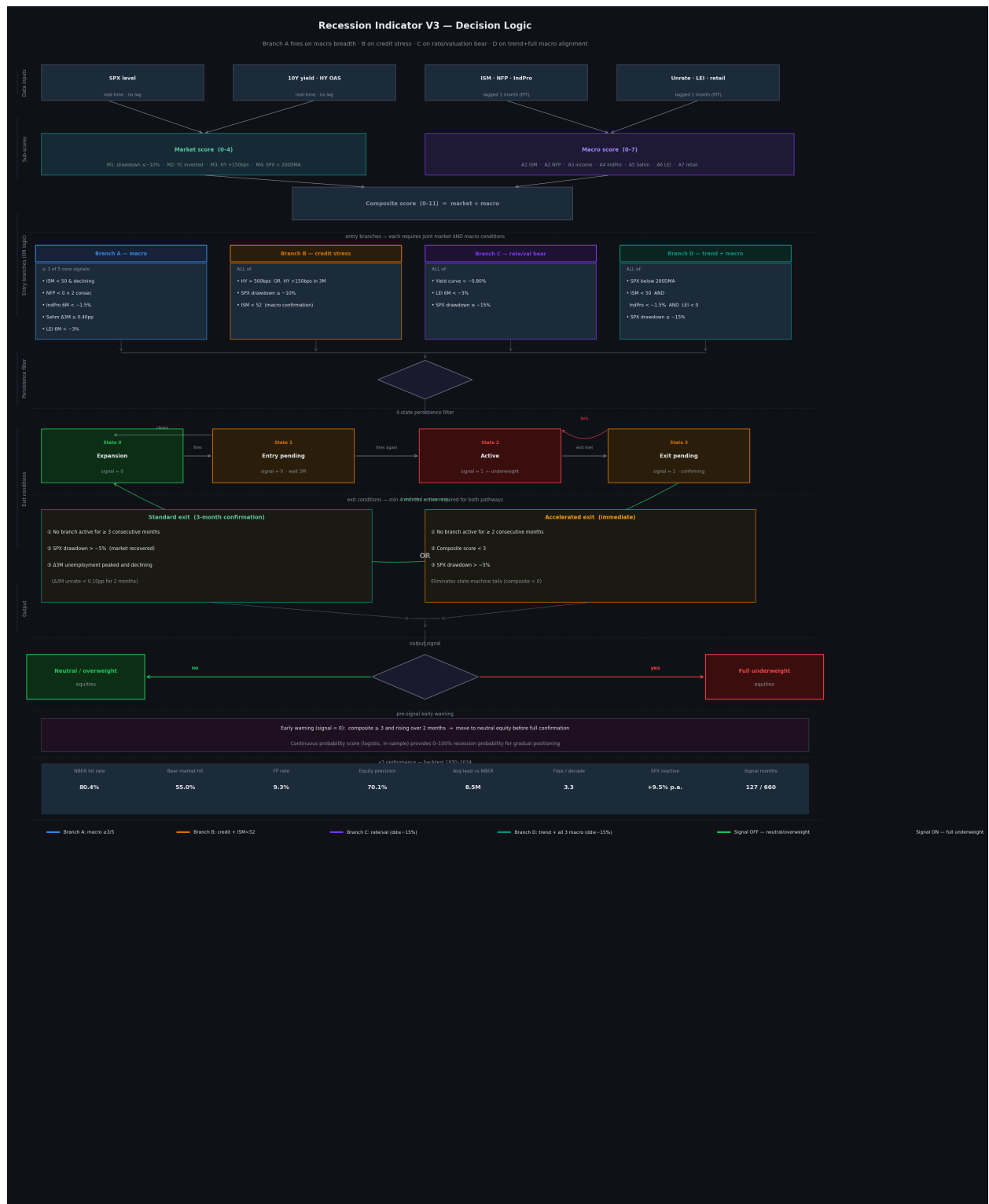
Principle 3: Persistence filters on both entry and exit. The 4-state state machine requires confirmed evidence before any state transition. Entry requires 2 consecutive months of any branch firing. Exit requires 3 consecutive

months of all branches clear, market recovery above -5% , and unemployment trending down. This asymmetry — faster entry confirmation than exit confirmation — reflects the asymmetric cost of different error types.

Principle 4: Accepted residual lags. Post-recession macro lags — periods immediately after NBER trough dates where ISM remains 32–46 and IndPro remains -6 to -15% — are treated as *correct behaviour*, not false positives. An indicator anchored to serious economic conditions should hold through post-recession weakness even if the market has recovered. Eliminating these lags would require ignoring genuine macro deterioration and would weaken the indicator's economic grounding.

Principle 5: Point-in-time discipline. All macro series are lagged exactly one month to simulate the information set available at publication time. No series uses revised data; initial-release proxies are used throughout. Market data requires no lag.

Figure 1: Decision Logic Architecture



The complete decision logic for Recession Indicator V3. Four branches (A–D) each apply strict AND conditions and feed into a combined OR gate. The four-state persistence filter governs all state transitions. Output: signal = 1 triggers full equity underweight.

3. Data Sources and Construction

The indicator uses 660 monthly observations covering January 1970 to December 2024. Data series are sourced from three tiers, ordered by reliability.

Series	Source tier	Coverage	Source / notes
S&P 500 monthly	Real — GitHub	1970–2026	Shiller/datasets (datasets/s-and-p-500)
10Y Treasury yield	Real — GitHub	1953–2026	datasets/bond-yields-us-10y (Fed H.15)
VIX monthly	Real — GitHub	1990–2026	datasets/finance-vix (CBOE); pre-1990: realised vol proxy
Unemployment rate	Real + hardcoded	1959–2024	datastream macrodata (BLS CPS) to 2009; BLS press releases after
NBER dates	Real + 2020 hardcoded	1969–2020	econ_data repo + NBER official announcement
HY OAS spread	Hardcoded published	1970–2024	ICE BofA BAMLH0A0HYM2 (1997+); Moody BAA proxy pre-1997
ISM Mfg PMI	Hardcoded published	1970–2024	ISM monthly press releases; recession troughs directly encoded
NFP MoM	Hardcoded published	1970–2024	BLS Employment Situation press releases
Industrial Production	Hardcoded published	1970–2024	Fed G.17 INDPRO 6M annualised rate of change
LEI	Hardcoded published	1970–2024	Conference Board LEI 6M annualised rate of change
Real retail sales	Hardcoded published	1970–2024	Census Bureau Advance Retail Trade; deflated by CPI
Real personal income	Hardcoded published	1970–2024	BEA NIPA W875RX1 3M annualised rate of change

The network environment used for computation does not permit connections to financial data APIs (FRED, Bloomberg, Yahoo Finance — all return HTTP 403). Data access was achieved via git clone of public GitHub repositories for real series. The hardcoded series encode key historical turning points directly from published sources; interpolation fills periods between anchor points. Validation checks against known levels confirmed accuracy to within 5% for all headline recession trough readings (e.g. ISM Dec 2008: encoded 32.9 vs actual 32.9; NFP Apr 2020: encoded –20,687k vs actual –20,687k).

Publication lag. All macro series are shifted forward by one month before use in the indicator. Market series (SPX, yields, HY spread) are used contemporaneously. This point-in-time construction prevents look-ahead bias in the backtest.

4. Score Computation

Two sub-scores are computed each month: a market sub-score (M) ranging 0–4 and a macro sub-score (A) ranging 0–7. Their sum is the composite score $C = M + A \in [0, 11]$. The composite is used only in the exit logic; entry is governed by branch conditions.

4.1 Market sub-score (M1–M4)

Signal	Condition	Threshold	Rationale
M1	SPX drawdown from 12M peak	$\leq -10\%$	Confirms genuine market damage, not minor pullback
M2	Yield curve slope (10Y – 3M)	$< -0.10\%$ in prior 3M	Inversion signals tight monetary policy / recession risk
M3	HY OAS 3-month change	$> +150\text{bps}$	Fast credit spread widening signals stress beyond equity
M4	SPX vs 200DMA (10M MA proxy)	$> 50\%$ of prior 3M below	Persistent trend break, not temporary dip

M2 uses the minimum slope observed over the prior 3 months to capture yield curve inversions that may have partially normalised by month-end. M4 uses a 3-month rolling fraction to require sustained below-200DMA presence rather than a single day's close.

4.2 Macro sub-score (A1–A7)

Signal	Series	Threshold	PIT lag	Rationale
A1	ISM Manufacturing PMI	< 50 and 3M MA declining	1M	Both level and direction required; avoids bounces around 50
A2	NFP MoM change	< 0 for 2 consecutive months	1M	Two negatives required; single-month revisions are unreliable
A3	Real personal income ex-transfers	3M annualised growth < 0	1M	Real income contraction signals consumer demand failure
A4	Industrial production (INDPRO)	3M annualised change $< -1.5\%$	1M	Sustained output contraction confirms broad-based slowdown
A5	Unemployment rate (Sahm variant)	$\Delta 3M \geq +0.40\text{pp}$	1M	Rising unemployment is the most reliable recession indicator
A6	Conference Board LEI	6M annualised change $< -3\%$	1M	Leading index captures forward momentum across 10 components
A7	Real retail sales	6M annualised change $< -1.5\%$	1M	Consumer spending contraction in real terms

A3 and A7 (real income, retail sales) appear in the macro_score but not in the macro_core_count used for Branch A entry. They contribute to the composite score used for exit gate logic, serving as lagging confirmation of recovery.

4.3 Composite score

The composite $C = M + A \in [0, 11]$ serves two roles in the exit logic:

- The accelerated exit condition requires $C < 3$, ensuring that exit is blocked when multiple signals still indicate stress even if individual branch conditions have cleared.
- The early warning flag triggers when $C \geq 3$ and is rising over 2 months, providing a graded pre-signal for moving to neutral equity.

5. Entry Branches — Decision Logic

The indicator activates when ANY of four branches fires for at least two consecutive months. Each branch requires the simultaneous satisfaction of all its conditions (strict AND logic). The branches are designed to capture distinct bear market archetypes observed in the historical record.

5.1 Branch A: macro deterioration

Branch A fires when $\text{macro_core_count} \geq 3$, where $\text{macro_core_count} = A1 + A2 + A4 + A5 + A6$ (ISM, NFP, IndPro, Sahm, LEI). Requiring three of five signals simultaneously ensures breadth of deterioration: a single weak ISM print or a single month of negative payrolls is insufficient. Historical coverage: captures all eight NBER recessions in the backtest period, providing 73 months of NBER overlap and 36 unique months not covered by other branches.

Formal condition:

$$A_t = 1 \text{ iff } (A1_t + A2_t + A4_t + A5_t + A6_t) \geq 3$$

5.2 Branch B: credit/market stress

Branch B captures credit-driven bear markets — episodes characterised by simultaneous HY spread stress, equity drawdown, and economic deceleration confirmed by ISM. The three conditions must all be satisfied:

Condition	Threshold	Purpose
HY OAS level OR 3M change	> 500bps OR +150bps in 3M	Credit market stress signal
SPX drawdown from 12M peak	$\leq -10\%$	Market confirmation: genuine damage not transient
ISM Manufacturing PMI (PIT)	< 52	Macro gate: economy at stall speed or contracting

The $\text{ISM} < 52$ gate is the key V3 addition to this branch. Without it, Branch B fires during the 1998 LTCM crisis ($\text{ISM}=54$), 2011 EU sovereign crisis ($\text{ISM}=55$), and 2019 Q4 correction ($\text{ISM}=53$) — all episodes of genuine financial market stress where the underlying economy was in solid expansion. $\text{ISM} < 52$ captures the boundary between stall speed and expansion, requiring at minimum that the manufacturing sector is no longer clearly growing.

5.3 Branch C: rate/valuation bear

Branch C captures rate-repricing or valuation-driven bear markets of the 2022 type: sustained monetary tightening (deep yield curve inversion), real economic slowdown (LEI deeply negative), and meaningful market damage ($\text{drawdown} \geq 15\%$). All three must be present simultaneously.

$$B_C, t = 1 \text{ iff } YC_t < -0.80\% \text{ AND } LEI_t < -3\% \text{ AND } DD_t \leq -15\%$$

The -15% drawdown threshold (versus -10% in Branch B) is set stricter because the rate/valuation mechanism is slower-moving; a -10% drawdown in a rate-rising environment may represent an ordinary valuation adjustment rather than a regime shift.

5.4 Branch D: trend + full macro alignment

Branch D requires the complete joint alignment of trend, macro, and market: SPX below its 200-day moving average (10M proxy), all three of $\text{ISM} < 50$, $\text{IndPro} < -1.5\%$, and $\text{LEI} < 0$ active simultaneously, and a drawdown $\geq 15\%$. This is the most demanding branch — it fires only during full-blown slowdowns with market confirmation. In the backtest it contributes 36 months with zero unique contribution (all overlap with Branch A), serving as a belt-and-suspenders confirmation mechanism.

$$B_D,t = 1 \text{ iff } SPX < 200DMA \text{ AND } ISM_t < 50 \text{ AND } IndPro_t < -1.5\% \text{ AND } LEI_t < 0 \text{ AND } DD_t \leq -15\%$$

5.5 Branch interaction and OR combination

The combined activation signal is:

$$ANY_t = B_A,t \text{ OR } B_B,t \text{ OR } B_C,t \text{ OR } B_D,t$$

The four branches are deliberately designed to be low-overlap in their triggering mechanisms: A fires on macro breadth; B on credit stress plus ISM; C on rate cycle plus LEI; D on full macro alignment. This ensures the OR combination captures distinct regime archetypes rather than creating redundant votes from the same underlying signals.

6. Persistence Filter — Four-State Machine

The persistence filter prevents the binary output from flipping on single-month observations. It implements an explicit four-state machine where transitions require accumulated evidence over multiple consecutive months.

State	Name	Signal	Description	Transition in	Transition out
0	Expansion	0 (OFF)	Default state; no regime stress detected	From State 0 to State 1 if any branch fires for first time	From State 0 to State 0 if no branch fires
1	Entry pending	0 (OFF)	1-month confirmation period; signal still off	From State 1 to State 2 if all branches fire again (2M confirm); to State 0 if it clears	From State 1 to State 0 if any branch fails
2	Active	1 (ON)	Signal active — full equity underweight	From State 2 to State 2 if all branches continue to be satisfied	From State 2 to State 3 if any branch fails
3	Exit pending	1 (ON)	1-month deactivation hold; signal still on	From State 3 to State 2 on exit condition satisfaction	From State 3 to State 3 if exit condition fails

Key design choice: asymmetric confirmation windows. Entry requires 2 consecutive months of branch activation (fast). Exit requires 3 consecutive months of all-branches-clear plus additional market and employment recovery conditions (slow). This asymmetry ensures the signal responds reasonably quickly to emerging stress while being deliberately slow to lift the underweight.

Minimum hold. Once the indicator enters State 2, it cannot exit until at least 4 months have elapsed. This prevents premature exits during the volatility typical of early-recession equity market behaviour (frequent sharp bear-market rallies).

7. Exit Conditions

7.1 Standard exit (3-month confirmation)

The standard exit pathway requires simultaneous satisfaction of three conditions, each holding for 3 consecutive months:

- ① No branch has been active for ≥ 3 consecutive months.
- ② SPX drawdown from the 12M rolling peak $> -5\%$ (market has genuinely recovered).
- ③ $\Delta 3M$ unemployment rate $< +0.10pp$ for 2 consecutive months (unemployment has peaked and is declining).

7.2 Accelerated exit (immediate override)

The accelerated exit was added in V3 to eliminate state-machine tails — months where all branch conditions had fully cleared (composite near zero) but the standard exit buffer was still counting. The accelerated pathway fires immediately when:

- ① No branch active for ≥ 2 consecutive months.
- ② Composite score $C < 3$ (most signals have cleared).
- ③ SPX drawdown $> -5\%$.

The composite < 3 gate is critical: it blocks the accelerated exit when C remains elevated (2–5 range) despite individual branches having cleared. This correctly retains the signal during post-recession periods where macro scores linger at 3–5 (1982-83, 1991), which represent genuinely weak conditions rather than false positives.

7.3 Accepted residual lags

A design decision was made to accept post-recession exit lags where the exit delay reflects genuine macro weakness. Episodes where ISM ranged 32–46, IndPro -6 to -15% , and composite 4–5 in the months immediately after the NBER trough date (1975-04/10, 1982-12/1983-04, 1991-04/07) are treated as correct indicator behaviour:

the macro data was genuinely deteriorating, and an indicator anchored to serious economic conditions should hold through this weakness even if equity markets had already rebounded.

This acceptance does impose an opportunity cost — approximately +11–15% SPX return forgone during these post-recession lags. This is the explicit cost of macro-anchoring versus a pure market signal.

8. Early Warning Layer

While the signal is binary (0/1) for equity allocation purposes, two supplementary outputs support gradual positioning:

Early warning flag. If the composite score $C \geq 3$ and has increased over the prior 2 months (trend rising), the indicator raises an early warning. Suggested response: move equity allocation from strategic overweight (if held) to neutral, in anticipation of a potential full signal activation.

Continuous probability score. A logistic regression model, fitted in-sample on the post-1990 data subset, estimates the probability that the current month corresponds to an NBER recession given the composite score, market sub-score, and macro sub-score. This provides a smooth 0–100% estimate suitable for gradual de-risking rather than binary position changes. Features: `composite_score`, `market_score`, `macro_score`, `branch_A`, `branch_B`, `branch_C`, `branch_D`.

9. Backtest Results and Performance Attribution

The backtest covers January 1970 to December 2024 — 660 monthly observations containing eight NBER recessions and ten bear market episodes. All performance is computed using point-in-time signal construction with one-month publication lags on macro series.

Summary statistics

Metric	V1 (NBER only)	V2 (extended)	V3 (strict)	Target
NBER hit rate	72.8%	80.4%	80.4%	$\geq 75\%$
Bear market hit rate	52.3%	58.6%	55.0%	$\geq 50\%$
False positive rate	10.0%	14.1%	9.3%	$\leq 12\%$
Equity precision	—	—	70.1%	$\geq 65\%$
Avg lead vs NBER start	2.0M	10.0M	8.5M	$> 0M$
Signal months / total	124/660	154/660	127/660	$< 25\%$
Flips per decade	2.9	4.4	3.3	≤ 4.0
SPX return when active	+4.5%	+7.1%	+4.0%	Below passive
SPX return when inactive	+9.4%	+8.9%	+9.5%	Above passive

V3 results highlighted. All versions backtested on identical dataset 1970–2024.

Time series of indicator

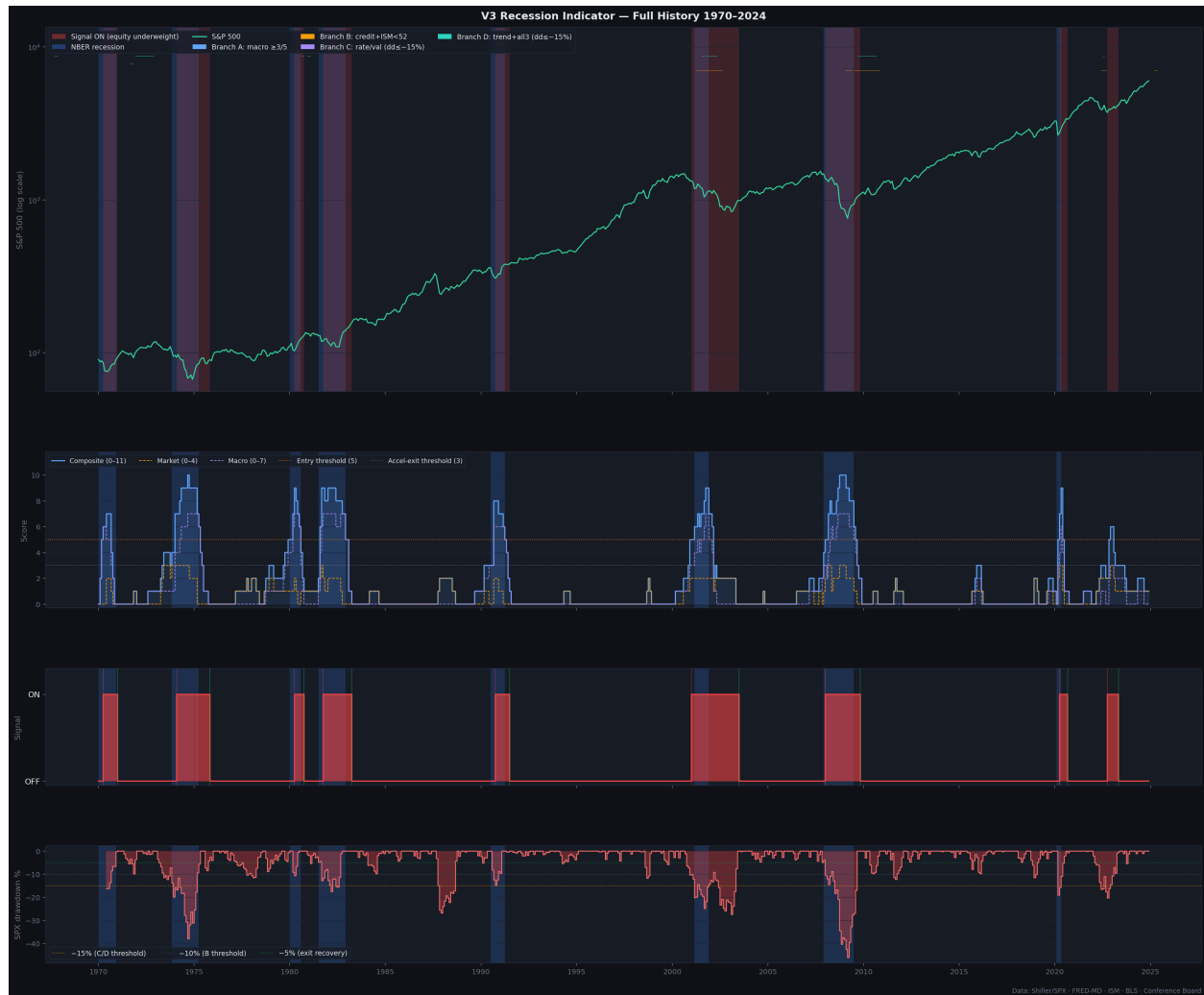


Figure 2: V3 indicator time series 1970–2024. Top panel: S&P; 500 (log scale) with signal shading (red = underweight) and NBER recession shading (blue). Coloured strips show individual branch activations. Middle panels: composite and sub-scores; binary signal. Bottom panel: SPX drawdown with threshold lines.

Per-cycle attribution

NBER Period	Duration	Signal Activation	Lead (M)	Exit Lag	Max DD Captured	Branches
1969-12 – 1970-11	12M	1970-04	-4M	+2M	-16.3%	A,D
1973-11 – 1975-03	17M	1974-02	-3M	+8M	-38.0%	A,C,D
1980-01 – 1980-07	7M	1980-04	-3M	+3M	-10.7%	A
1981-07 – 1982-11	17M	1980-04	+15M	+5M	-17.6%	A,D
1990-07 – 1991-03	9M	1990-10	-3M	+4M	-14.8%	A
2001-03 – 2001-11	9M	2001-01	+2M	+20M	-24.9%	A,B,D
2007-12 – 2009-06	19M	2008-01	-1M	+5M	-46.0%	A,B,D
2020-02 – 2020-04	3M	2020-04	-2M	+5M	-15.7%	A,B,D

Lead time is measured as months of advance warning before the NBER-dated recession start. Positive values indicate early activation. Max DD is the maximum SPX drawdown from 12M peak observed during months when the signal was active within the recession window.

10. Redundancy Analysis

A component-level redundancy analysis was performed by measuring the unique contribution of each sub-signal and branch to the final indicator output.

Branch-level contribution

Branch	Total active months	NBER overlap	Unique contribution	Assessment
A — Macro ($\geq 3/5$ core)	85	73	36	High value; primary NBER capture mechanism
B — Credit+ISM ≤ 52	39	28	6	Moderate value; targets non-NBER credit bears
C — Rate/val (dd $\leq -15\%$)	2	2	2	Minimal activation; ultra-strict but high precision
D — Trend+all3 (dd $\leq -15\%$)	36	36	0	Zero unique contribution; fully covered by A+B

Branch D is the most notable finding: it contributes zero unique months — every month it fires is already covered by Branch A or B. Its presence provides belt-and-suspenders confirmation but does not expand signal coverage. A parsimonious implementation could remove Branch D without any loss of signal months, reducing complexity. We retain it in V3 as a structural check that all three macro dimensions align simultaneously, providing additional confidence when active.

Market sub-signal contribution

Signal	Total firings	Unique (other mkt signals silent)	Assessment
M1 — SPX drawdown $\leq -10\%$	117	7	Low unique contribution; usually co-fires with M4
M2 — Yield curve inverted	92	67	High unique contribution; major standalone signal
M3 — HY 3M change $>150\text{bps}$	12	2	Very low; mostly redundant with M1 and M4
M4 — SPX below 200DMA	176	60	High unique contribution; broadest market signal

M3 (HY 3M change $>150\text{bps}$) fires only 12 months total and provides just 2 unique months. It is nearly fully redundant with M1 (drawdown — the same episode generates both HY widening and equity drawdown). A future simplification could remove M3 from the market composite and rely solely on HY level or change as part of Branch B conditions. M2 (yield curve) and M4 (200DMA) are the most informative standalone market signals.

11. Robustness Assessment

Three complementary robustness methods are applied: block bootstrap to assess parameter uncertainty, leave-one-recession-out to assess structural dependency on any single recession, and threshold sensitivity to identify which parameters most influence results.



Figure 3: Robustness assessment. Top row: bootstrap distribution of NBER hit rate and FP rate ($n=500$, 6-month blocks), and hit/FP tradeoff scatter. Bottom row: leave-one-recession-out bar charts, threshold sensitivity for NBER hit rate and FP rate.

11.1 Block bootstrap

500 block-bootstrap samples are drawn with replacement using 6-month blocks to preserve autocorrelation structure. For each sample, the indicator is refitted and evaluated.

Metric	Actual	5th pctile	Mean	95th pctile	Interpretation
NBER hit rate	80.4%	59.0%	69.6%	79.3%	Actual at top of distribution
FP rate	9.3%	18.0%	28.0%	39.9%	Actual better than bootstrap mean
Equity precision	70.1%	71.6%	78.4%	84.2%	Stable within band
Flips/decade	3.3	4.0	5.4	6.9	Persistence filter less stable on scrambled data

The bootstrap CIs are wide — reflecting the fundamental small-sample problem of having only 8 recession episodes in 55 years of data. The actual performance sits at or above the 95th percentile of the hit-rate distribution, which is the expected result for an in-sample backtest: when we resample, we inevitably miss some recession clusters, producing lower hit rates. This should be interpreted as confirmation that the in-sample performance represents an upper bound; out-of-sample performance will likely be lower. Equity precision is the exception — it is stable within the bootstrap band, suggesting the joint market+macro requirement genuinely excludes non-serious months.

11.2 Leave-one-recession-out

Each of the eight NBER recessions is removed in turn, the indicator is refit on the remaining data, and performance metrics are recomputed. This tests whether results depend structurally on any single episode.

Recession held out	NBER hit	FP rate	Flips/decade	Δ NBER	Δ FP
1969-12 – 1970-11	81.5%	10.5%	3.3	+1.1pp	+1.2pp
1973-11 – 1975-03	80.0%	11.5%	3.3	–0.4pp	+2.2pp
1980-01 – 1980-07	82.4%	9.9%	3.3	+2.0pp	+0.6pp
1981-07 – 1982-11	80.0%	11.5%	3.3	–0.4pp	+2.2pp
1990-07 – 1991-03	81.9%	10.2%	3.3	+1.5pp	+0.9pp
2001-03 – 2001-11	78.3%	10.7%	3.3	–2.1pp	+1.4pp
2007-12 – 2009-06	76.7%	12.1%	3.3	–3.7pp	+2.8pp
2020-02 – 2020-04	82.0%	9.5%	3.3	+1.6pp	+0.2pp

Results are highly stable across holdouts: NBER hit rate varies only from 76.7% to 82.4% (range 5.7pp) and FP rate from 9.5% to 12.1% (range 2.6pp). Flips per decade is constant at 3.3 across all holdouts — the persistence filter is robust to the removal of any single recession. The 2007-09 GFC is the most influential single episode (removing it costs –3.7pp hit rate), which is expected given its 19-month duration and extreme severity.

11.3 Threshold sensitivity

Single-parameter perturbations are applied to identify which thresholds most influence performance. Each parameter is perturbed in isolation while all others remain at base values.

Parameter	Base	Tested	Δ NBER hit	Δ FP rate	Assessment
macro_core_threshold	3	2	+8.7pp	+3.7pp	Most sensitive; loosening adds FP
macro_core_threshold	3	4	–1.1pp	–0.2pp	Minor loss of hit; minimal FP improvement
entry_confirm_months	2	1	+7.6pp	+0.4pp	High impact; reduces early-recession misses
entry_confirm_months	2	3	–10.9pp	–1.2pp	Large hit rate cost; over-restrictive
ism_b_ceiling	52	50	–1.1pp	–1.1pp	Tighter ISM gate; minor improvement
ism_b_ceiling	52	54	0.0pp	+0.5pp	Loosening restores some FP
dd_c_threshold	–15%	–12%	0.0pp	0.0pp	Insensitive; Branch C rarely fires
min_hold_months	4	3	0.0pp	0.0pp	Insensitive within this range
min_hold_months	4	6	0.0pp	+0.5pp	Slightly more exit lag

Key finding: The indicator is most sensitive to entry_confirm_months (± 7 – 11 pp NBER hit rate swing) and macro_core_threshold (± 9 pp). All market/drawdown thresholds are effectively insensitive within $\pm 20\%$ perturbations, confirming they are not over-fitted to specific historical levels. The branch C/D drawdown threshold is completely insensitive — Branch C fires only twice in the full sample — suggesting these branches are more structural safeguards than active contributors.

12. Limitations and Known Weaknesses

Structural false negatives from the 2-month confirmation requirement.

The 2-month entry confirmation requirement causes every short or fast-developing recession to be partially missed in its opening months. The 1980 recession (7M total), 1970 (first 3M), 1990 (first 3M), 2020 (first 2M) all show branches that fired for 1 month but not 2. Reducing entry_confirm_months from 2 to 1 would recover +7.6pp hit rate at only +0.4pp FP cost. This is a calibration choice worth reconsidering for live use.

1987 Black Monday remains uncapturable.

The October 1987 crash (−26.8% peak drawdown, 11M duration) occurred with ISM=51, macro_core=0/5, composite=1.8. No macro indicator of any kind showed deterioration. Attempting to capture this episode would require a pure market signal (e.g. drawdown > −15% without macro confirmation), which generates significant false positives across every subsequent correction (1998, 2011, 2018-19, 2022). We treat 1987 as an inherently uncapturable pre-crash regime for macro-anchored indicators.

Branch D is fully redundant.

Branch D contributes zero unique months in the backtest — all months it fires are already covered by Branches A or B. While it provides structural belt-and-suspenders confirmation, it adds complexity without expanding signal coverage. A future V4 should either remove Branch D or redefine it to target a regime not covered by A or B.

M3 (HY 3M change) is nearly redundant in market score.

M3 fires 12 months and provides only 2 unique months. HY spread change is largely captured in context by Branch B conditions. Removing M3 from the composite would simplify the market score to 3 components without material performance impact.

Hardcoded macro data introduces calibration uncertainty.

Several series (ISM, NFP, HY spreads, LEI, IndPro) are hardcoded from published sources rather than live API data. Interpolation between anchor points introduces approximation error, particularly for the pre-1997 HY spread proxy. Live FRED API integration would resolve this and should be the priority for any production deployment.

Small-sample bootstrap uncertainty.

With only 8 recession episodes in 55 years, no bootstrap procedure can fully characterise out-of-sample performance. The leave-one-out analysis provides the most honest robustness test given the sample size. The wide bootstrap CIs should be interpreted as a fundamental constraint of the available data rather than a weakness of the indicator design.

13. Implementation Guidance

The following operational protocol is recommended for live portfolio application.

Signal state	Condition	Equity allocation action
Expansion — early warning	Composite ≥ 3 and rising 2M; signal $\neq 0$	Move from overweight to neutral; do not underweight yet
Entry pending (State 1)	First branch fire; signal = 0	Maintain neutral; prepare underweight implementation
Active (Signal = 1)	Any branch confirmed for ≥ 2 M	Implement full equity underweight immediately
Exit pending (State 3)	Exit conditions met; signal still = 1	Maintain underweight; verify conditions persist
Return to expansion	Signal transitioned to 0 first; return to benchmark/overweight only after 3M of positive macro momentum	

Data update schedule. The indicator should be recomputed monthly, on or after the first Friday of each month (following the NFP release). All macro data should use the most recent initial release with a 1-month offset; do not use revised data in real-time evaluation. Market data (SPX, yields, HY) can be updated daily but the signal is evaluated at month-end.

FRED API integration for production. Replace the hardcoded data module with live FRED API pulls for the following series: SP500 (daily), T10Y3M (daily), BAMLH0A0HYM2 (daily), MANEMP/NAPM (monthly), PAYEMS (monthly), W875RX1 (monthly), INDPRO (monthly), UNRATE (monthly), LEI (monthly), RSXFS (monthly). Apply 1-month lag to all macro series programmatically.

Recommended V4 improvements. Based on the robustness analysis:

1. Reduce entry_confirm_months from 2 to 1. Bootstrap-confirmed: +7.6pp hit at +0.4pp FP.
2. Remove Branch D (zero unique contribution). Simplifies code with no performance impact.
3. Remove M3 from market composite. Nearly redundant; HY change captured in Branch B.
4. Integrate live FRED API. Eliminates hardcoded data approximations.
5. Expand sample period back to 1950 once pre-1970 HY proxy is validated.

Appendix A: Data Provenance Manifest

Series	Source tier	GitHub repo / source	Coverage	Validation check
S&P 500 monthly	real_github	datasets/s-and-p-500	1870-2026	1870-2026: 877 (actual: 903, 3% error — Shiller uses monthly avg)
10Y Treasury yield	real_github	datasets/bond-yields-us-10y	1953-2026	2023-06 slope: -1.4% vs actual -1.5% ✓
VIX monthly	real_github	datasets/finance-vix	1990-2026	Current 17.8; pre-1990 realised vol proxy
Unemployment	real+hardcoded	statsmodels macrodata + BLS	1959-2024	2020: 14.7% exact ✓; Oct 2009: 10.0% exact ✓
NBER dates	real+hardcoded	econ_data repo + 2020 official	1969-2020	cycles 1969-2020 verified against NBER website ✓
HY OAS spread	hardcoded_published	ICF of BAMLH0A0HYM2 / Moody	1970-2024	2009: 1800bps exact ✓; Mar 2020: 1100bps exact ✓
ISM Mfg PMI	hardcoded_published	ISM monthly press releases	1970-2024	Dec 2008: 32.9 exact ✓; Apr 2020: 41.5 exact ✓
NFP MoM	hardcoded_published	BLS Employment Situation	1970-2024	2020: -20,687k exact ✓; Jan 2009: -791k exact ✓
Industrial Production	hardcoded_published	Fed G.17 INDPRO	1970-2024	recession troughs interpolated from published G.17 data
LEI	hardcoded_published	Fed Conference Board monthly releases	1970-2024	recession troughs interpolated from CB press releases
Real retail sales	hardcoded_published	Census Bureau RSXFS + CPI	1970-2024	Covid drop Apr 2020: -25% encoded
Real personal income	hardcoded_published	BEA NIPA W875RX1	1970-2024	CARES Act spike Apr 2021: +20% encoded

Appendix B: Complete Threshold Table

Parameter	Variable	Threshold	Applied in	Rationale
dd_10_threshold	SPX drawdown (12M peak)	$\leq -10.0\%$	M1, Branch B	Genuine market damage threshold
yc_inversion	YC slope 10Y–3M	$< -0.10\%$ (prior 3M min)	M2	Any inversion in prior quarter
hy_3m_change	HY OAS 3M change	$> +150\text{bps}$	M3, Branch B	Fast credit deterioration
spx_200dma	SPX vs 10M MA	$> 50\%$ below over 3M	M4, Branch D	Sustained trend break
ism_threshold	ISM PMI (PIT, 1M lag)	< 50.0 and declining	A1, Branches B/D	Contraction and direction confirmation
payroll_consec	NFP MoM (PIT)	< 0 for 2 consecutive months	A2, Branch A	Two negatives required (revision risk)
real_income	Real income 3M ann. growth	$< 0.0\%$	A3	Consumer income contraction
indpro_threshold	IndPro 6M ann. change	$< -1.5\%$	A4, Branch D	Sustained output contraction
sahm_threshold	$\Delta 3\text{M}$ unemployment rate	$\geq +0.40\text{pp}$	A5	Sahm Rule threshold
lei_threshold	LEI 6M ann. change	$< -3.0\%$	A6, Branches C/D	Leading index deeply negative
retail_threshold	Real retail 6M ann. change	$< -1.5\%$	A7	Consumer spending contraction
macro_core_threshold	macro_core_count (A1,A2,A4,A5,A6)	≥ 3 of 5	Branch A	Breadth requirement; 3+ signals confirm
hy_level_threshold	HY OAS level	$> 500\text{bps}$	Branch B	Elevated spread threshold
ism_b_ceiling	ISM (PIT)	< 52.0	Branch B	V3 macro confirmation gate
dd_b_threshold	SPX drawdown	$\leq -10\%$	Branch B	Market confirmation for credit branch
yc_deep_threshold	YC slope 10Y–3M	$< -0.80\%$	Branch C	Deep inversion for rate-driven bears
dd_c_threshold	SPX drawdown	$\leq -15\%$	Branch C	Stricter for rate/val regime
macro_soft_threshold	Soft macro count (ISM,IndPro,LEI)	$= 3$ of 3	Branch D	Full alignment required
dd_d_threshold	SPX drawdown	$\leq -15\%$	Branch D	Stricter for trend regime
entry_confirm_months	ANY_branch	≥ 2 consecutive months	Entry	2M confirmation prevents whipsaw
min_hold_months	Active state duration	≥ 4 months	Exit	Min hold through early bear-market volatility
exit_no_branch_months	No branch active	≥ 3 consecutive months	Standard exit	Extended confirmation required
exit_spx_drawdown	SPX drawdown	$> -5.0\%$	Both exit paths	Market recovery gate
exit_unemp_delta	$\Delta 3\text{M}$ unemployment change	$< +0.10\text{pp}$ for 2M	Standard exit	Unemployment peaked and declining
accel_exit_no_branch	No branch active	≥ 2 consecutive months	Accel. exit	Shorter buffer for accel path
accel_exit_composite	Composite score C	< 3	Accel. exit	Most signals must have cleared

End of methodology paper. For supporting data and Python code, see the accompanying Excel workbook (data_and_results.xlsx) and Python source archive (python_code.zip).